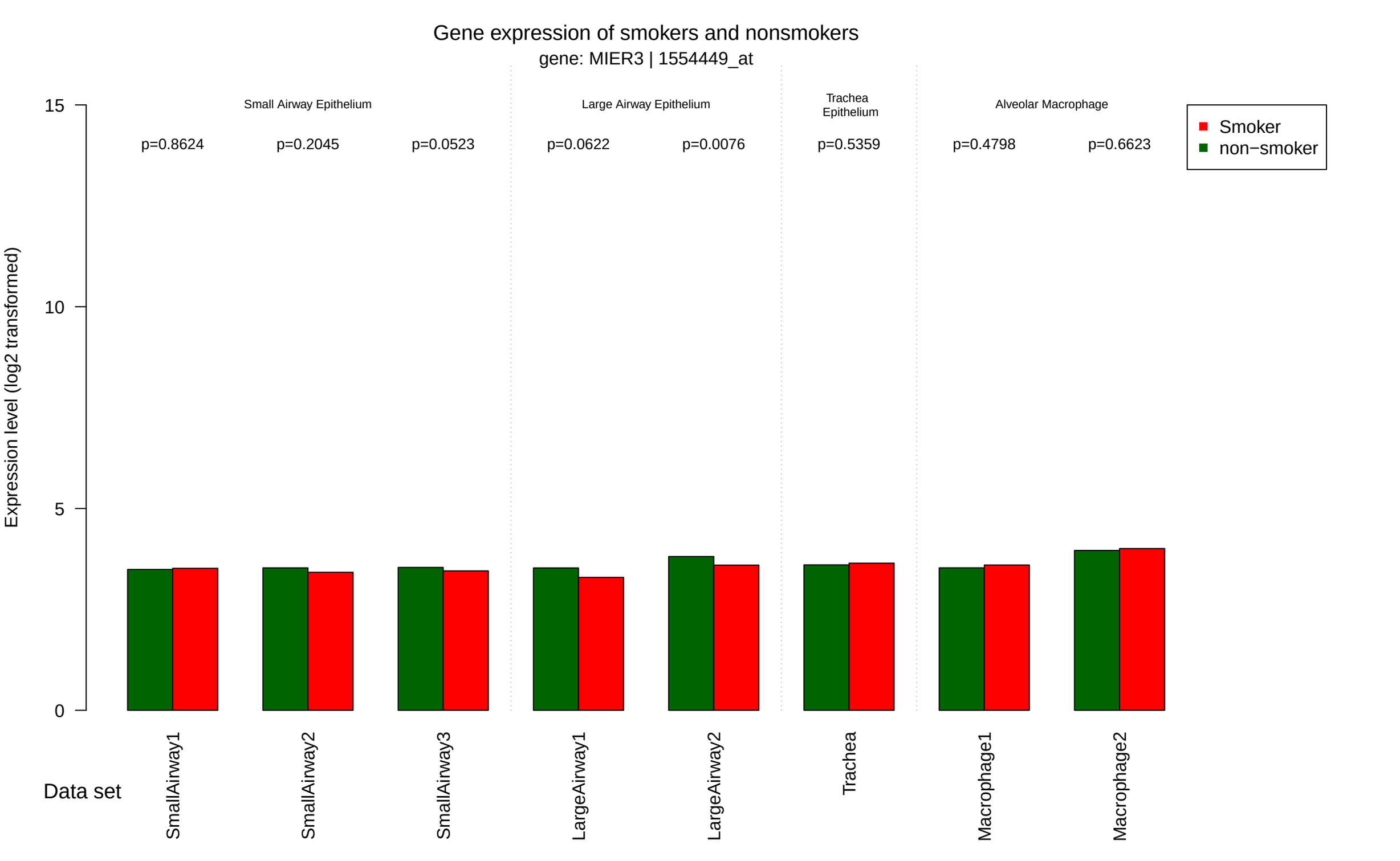
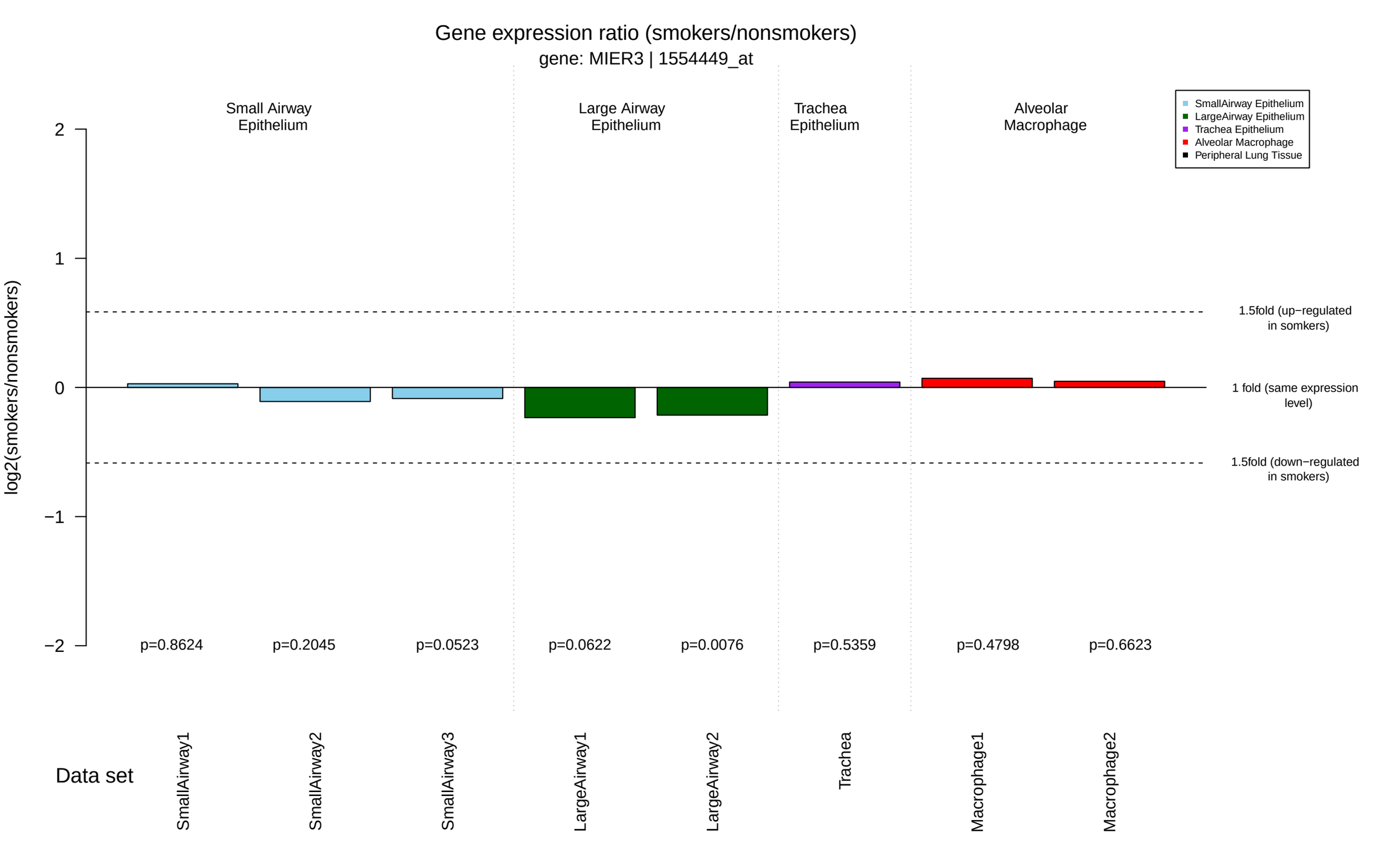
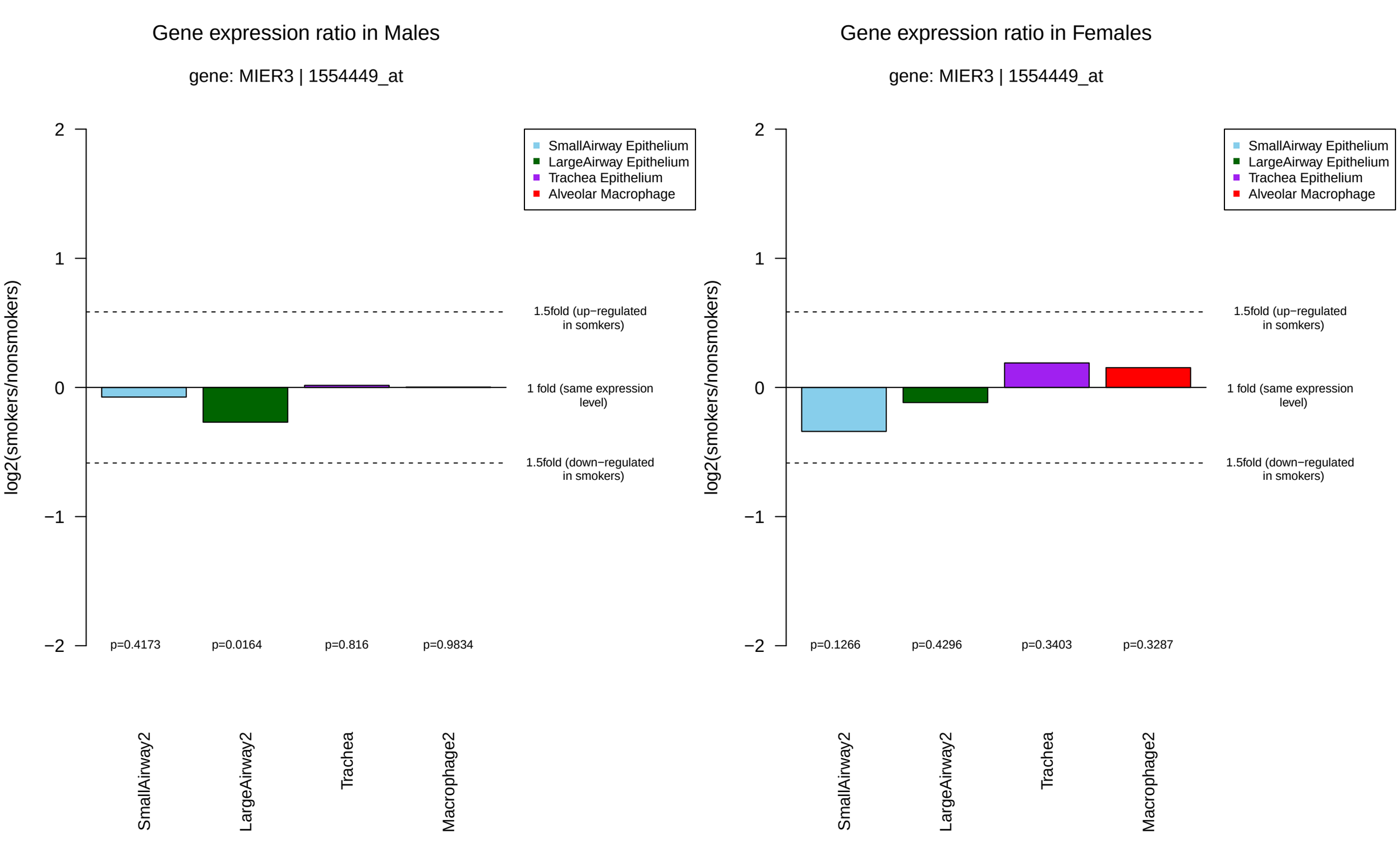
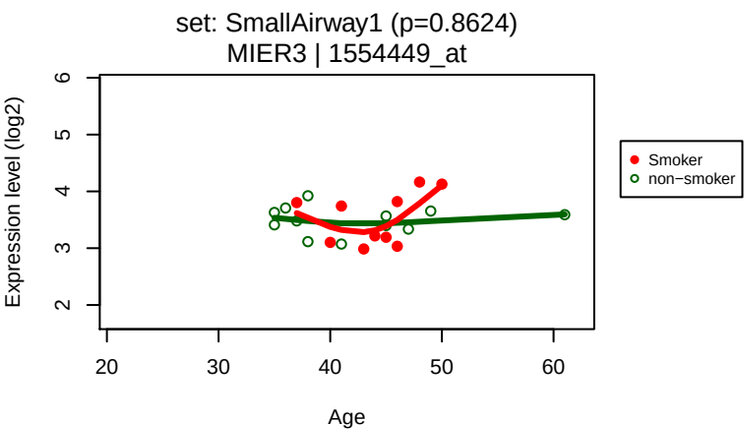
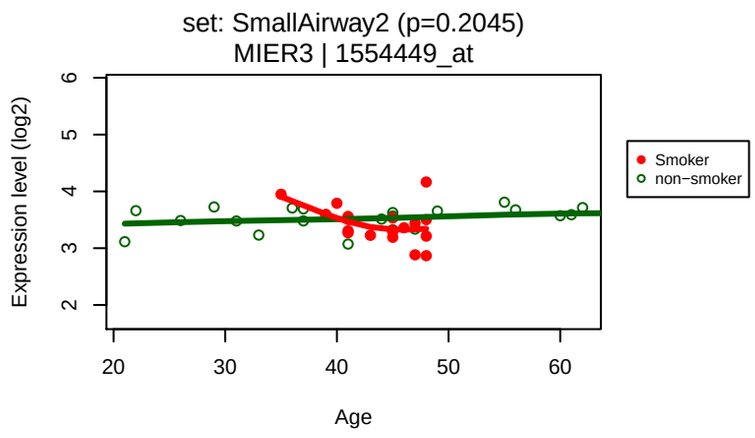
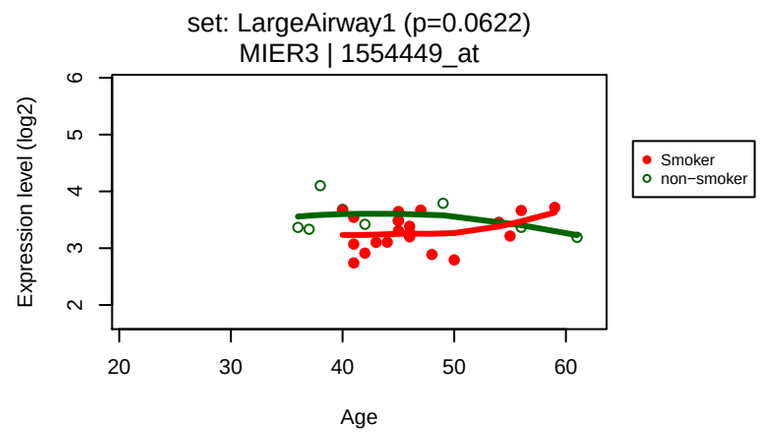

Gene expression of smokers and nonsmokers
gene: MIER3 | 1554449_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8624
p=0.2045
p=0.0523
p=0.0622
p=0.0076
p=0.5359
p=0.4798
p=0.6623
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

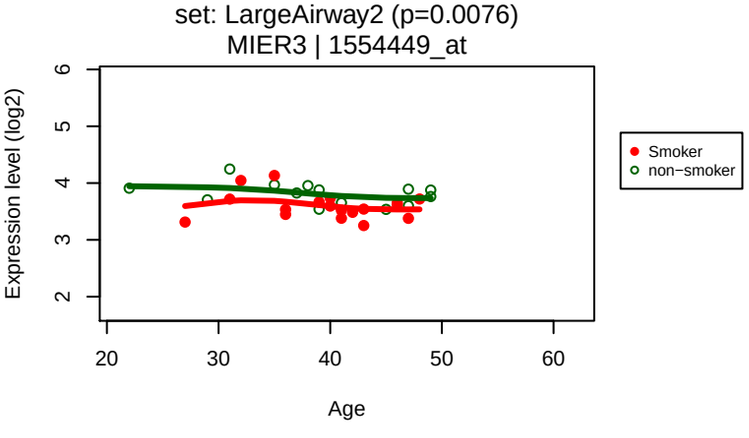

Gene expression ratio (smokers/nonsmokers)
gene: MIER3 | 1554449_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8624
p=0.2045
p=0.0523
p=0.0622
p=0.0076
p=0.5359
p=0.4798
p=0.6623
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

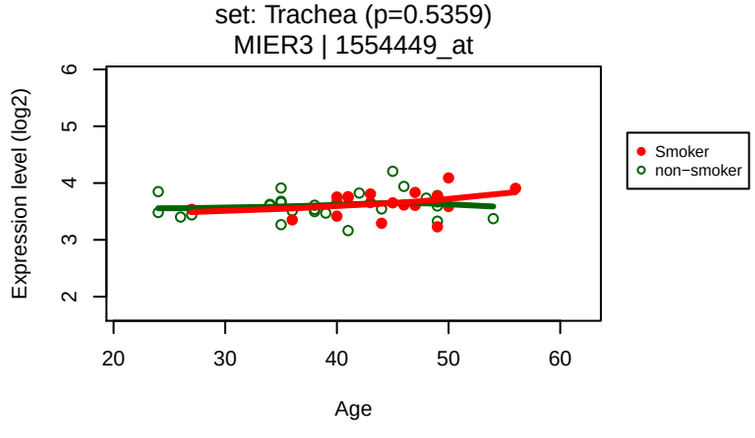

Gene expression ratio in Males
gene: MIER3 | 1554449_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4173
p=0.0164
p=0.816
p=0.9834
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MIER3 | 1554449_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1266
p=0.4296
p=0.3403
p=0.3287
SmallAirway2
LargeAirway2
Trachea
Macrophage2

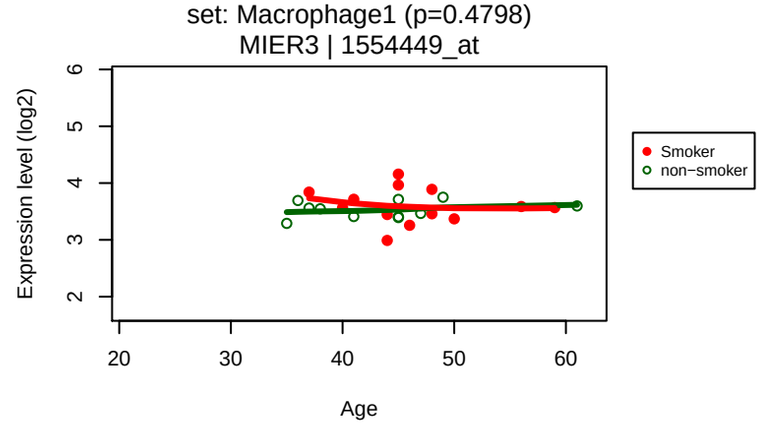

set: SmallAirway1 (p=0.8624)
MIER3 | 1554449_at
Expression level (log2)
Age
Smoker
non−smoker

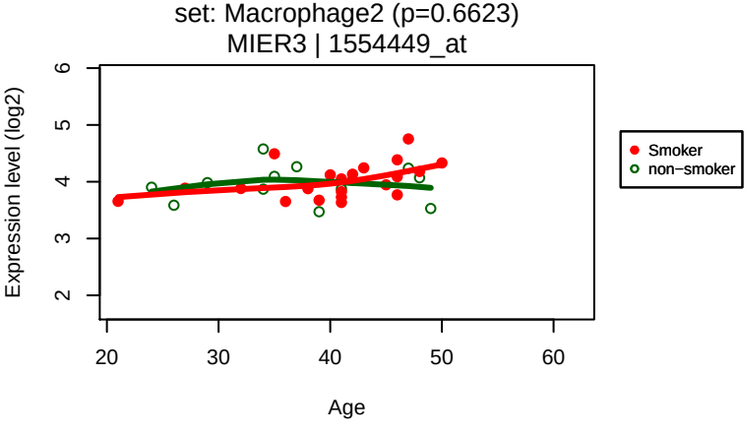

set: SmallAirway2 (p=0.2045)
MIER3 | 1554449_at
Expression level (log2)
Age
Smoker
non−smoker

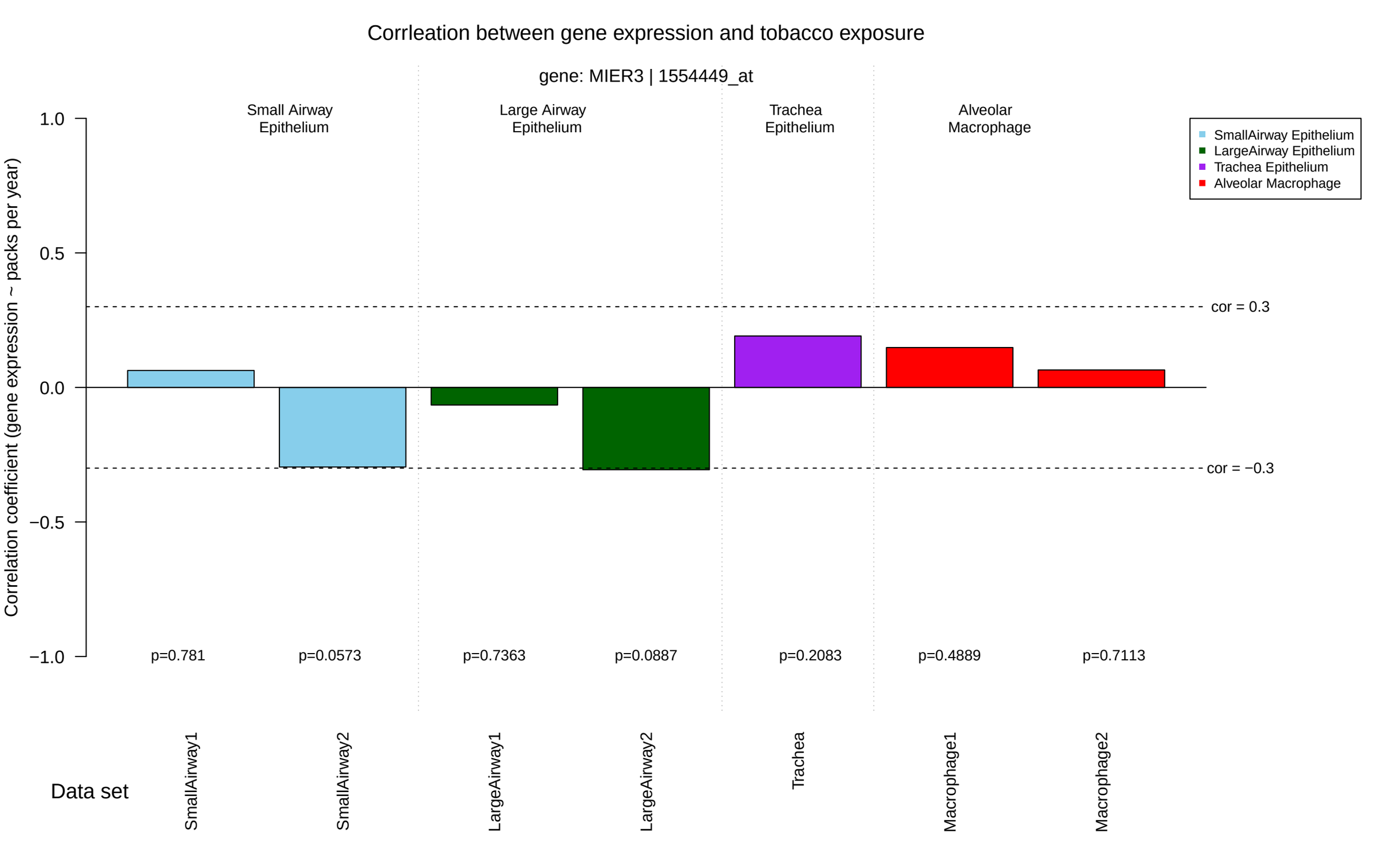

set: LargeAirway1 (p=0.0622)
MIER3 | 1554449_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0076)
MIER3 | 1554449_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5359)
MIER3 | 1554449_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4798)
MIER3 | 1554449_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6623)
MIER3 | 1554449_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MIER3 | 1554449_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.781
p=0.0573
p=0.7363
p=0.0887
p=0.2083
p=0.4889
p=0.7113
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2